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Evaluation of WRF Meteorology for a Uinta Basin Winter Ozone Episode

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1. Introduction

Winter ozone episodes in Utah's Uinta Basin require both rapid photochemical production and sustained meteorological confinement. A snow-covered surface supports the radiative environment for ozone formation, while a shallow cold-air pool limits dilution and allows precursors and ozone to accumulate near the basin floor [1,2]. Meteorological errors therefore propagate directly into any subsequent interpretation of simulated O_3 production, transport, and loss.

Previous coupled modeling studies have often attributed underestimated winter ozone to emissions inventories or chemical processes. Increasing oil-and-gas volatile-organic-compound emissions improved simulated ozone in both WRF–Chem and CMAQ studies [3,4]. Those experiments also exposed important meteorological and process uncertainties, including mixing-depth errors, snow-dependent photolysis and deposition, and compensating effects between transport and emissions. Their results establish that inventory uncertainty matters, but do not make meteorological fidelity optional.

Meteorology-focused studies demonstrate that Uinta Basin cold-air-pool simulations are sensitive to snow and cloud treatment, boundary-layer behavior, initialization, and data assimilation [2,5]. These sensitivities complicate attempts to identify a single best WRF configuration and can change the atmospheric volume into which chemical emissions are mixed. The meteorological component should consequently be evaluated on its own terms before it is used to diagnose chemical-model performance.

This study evaluates WRF simulations of the cold-air pool on 2 February 2013, during a documented Uinta Basin winter ozone episode. The analysis is designed to identify which configuration choices materially affect the simulated near-surface and lower-tropospheric structure, and to define a defensible meteorological basis for later coupled air-chemistry experiments.

2. Materials and Methods

2.1. Case, Observations, and Evaluation Sample

The model evaluation focused on 2 February 2013 from 14:00 to 18:00 UTC. The first two forecast hours were excluded as spin-up, leaving five common hourly samples. The observation record used eleven stations for 2 m temperature, ten for 10 m wind speed, and nine for surface pressure. The 1 km second domain was used for the common station evaluation because the 333 m third domain excludes most of the higher bench

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sites. Observed context from 31 January through 4 February was retained to show the floor-to-bench temperature evolution surrounding the evaluation window.

The horizontal bulk cold-pool contrast was calculated as the potential-temperature difference between Mountain Home on the northwest bench and Pariette Draw on the basin floor. This two-station contrast is a surface-based structural indicator rather than a measured free-air inversion depth.

2.2. WRF Experiment Matrix

The promoted simulations used WRF version 4.8.0 with three domains at 3, 1, and 0.333 km grid spacing, 75 vertical levels, and a 50 hPa model top. They used Thompson microphysics, RRTMG longwave and shortwave radiation, the revised MM5 surface layer, the Noah land-surface model, and the YSU planetary-boundary-layer scheme. All promoted simulations were free forecasts without FDDA and ran from 12:00 to 18:00 UTC.

The primary comparison changed the driving analysis from the NAM analysis product to the GFS analysis product while retaining the same grid, physics, two-way nest feedback, terrain source, initialization time, and valid window. The source-dependent metgrid level count was 40 for NAM and 27 for GFS. The secondary comparison used the same NAM configuration and changed only nest feedback from two-way to one-way.

2.3. Surface, Retention, and Ventilation Diagnostics

Surface temperature, earth-relative 10 m wind, and surface pressure were sampled at the nearest d02 grid cell to each reporting station. For the lower-tropospheric analysis, potential temperature and earth-relative wind were linearly interpolated to common height-above-ground coordinates at each station column. Fixed-layer stability was defined as potential temperature at 500 m AGL minus potential temperature at 50 m AGL. The ventilation proxy was the magnitude of the height-weighted mean horizontal wind vector from 50 to 500 m AGL. This fixed-layer transport wind is not a direct measure of basin export or a chemistry-model ventilation coefficient. Native WRF PBL height was retained as a scheme-dependent companion diagnostic rather than treated as an observed physical layer depth.

2.4. Dependence and Short-Sample Uncertainty

Run differences were paired by station and valid hour before spatial aggregation. The fixed station field was reduced to one spatial median per hour so that the stations were not treated as independent replicates. A two-hour circular moving-block resample of the five hourly medians was repeated 20,000 times, and the 2.5th–97.5th percentile range was retained as a descriptive sensitivity interval. This interval is not a formal confidence interval, and the five-hour experiment cannot establish multi-day cold-pool persistence.

2.5. Use of Artificial-Intelligence-Assisted Tools

During preparation of this study, the authors used Anthropic Claude and OpenAI Codex to assist with literature synthesis, provenance review, analysis-code review, and manuscript drafting and editing. The authors reviewed and edited all outputs and take full responsibility for the content of this publication.

3. Results

3.1. Observed Cold-Pool Context

The basin-floor station repeatedly cooled below the higher bench stations during the multi-day observation window, with the strongest separation occurring during local

nighttime and morning periods (Figure 1). The station record therefore supports the presence of a strong but evolving horizontal cold-pool contrast during the modeled episode.

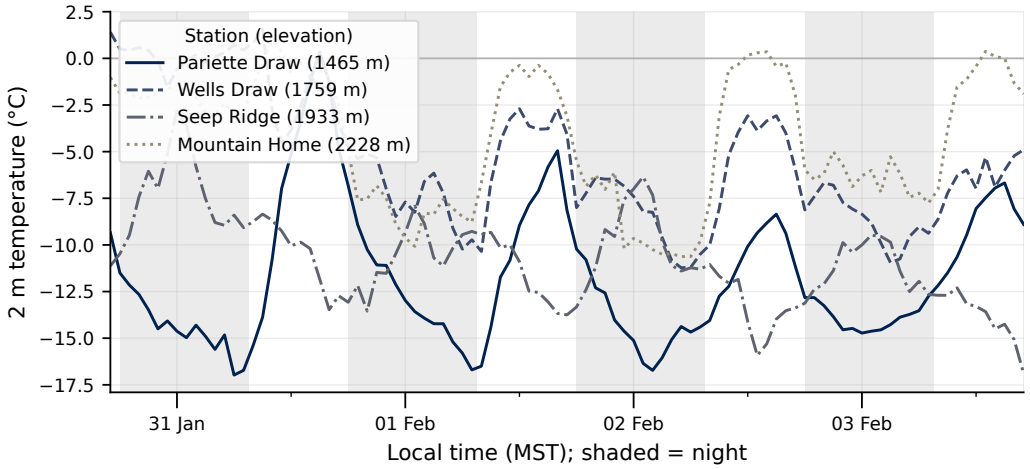


Figure 1. Observed 2 m-temperature evolution at four stations spanning the Uinta Basin floor and higher benches. The shaded intervals indicate local night. The sites provide elevation-spanning context but do not measure the full basin volume.

3.2. Response to Driving Analysis

The GFS-driven run produced a weaker horizontal bench-minus-floor potential-temperature contrast than the otherwise matched NAM-driven run throughout the evaluation window (Figure 2). The observed contrast ranged from 12.63 to 20.21 K, the NAM run ranged from 16.44 to 18.07 K, and the GFS run ranged from 7.35 to 11.74 K. These values describe a horizontal station contrast and are not interpreted as free-air inversion depths.

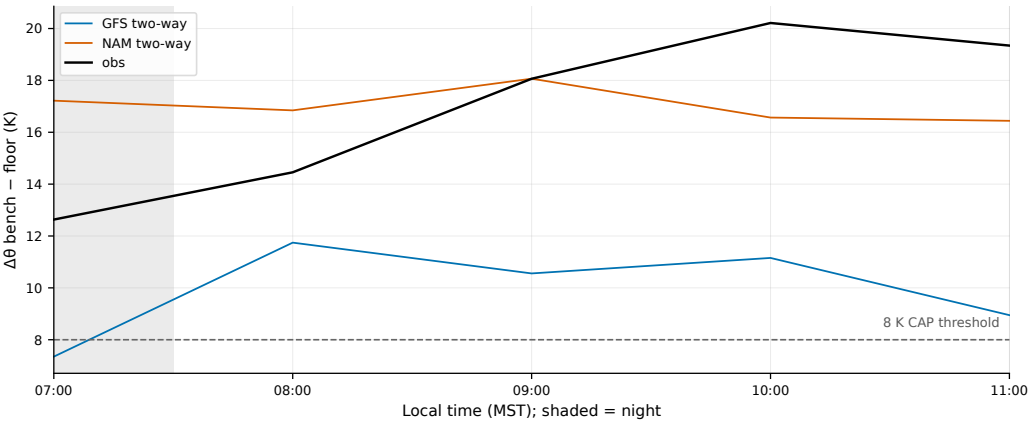


Figure 2. Observed and modeled Mountain Home-minus-Pariette Draw potential-temperature contrast during the matched evaluation window. The 8 K line is a literature yardstick rather than a case-specific physical threshold.

The mean hourly station-median 2 m-temperature bias was +2.09 °C for GFS and +0.40 °C for NAM. The paired GFS-minus-NAM bias difference was +1.79 °C, with a two-hour block sensitivity interval of +0.57 to +3.08 °C. The corresponding absolute-temperature-error difference was +1.41 °C, with an interval of +0.64 to +2.17 °C. The 10 m-wind bias was also larger in GFS by 0.91 m s^{-1}, although the difference in absolute wind error was not separated from zero by the block sensitivity interval.

The common-AGL profiles showed the same structural separation (Figure 3). Across the five hourly spatial medians, GFS-minus-NAM 50–500 m stability was -3.63 K, with an

interval of -4.94 to -2.22 K and the same sign in all five hours. GFS-minus-NAM transport wind was +0.79 m s^{-1}, with an interval of +0.58 to +0.99 m s^{-1} and the same sign in all five hours. The native PBL-height difference changed sign across the window and had an interval of -11.24 to +10.48 m, so PBLH alone did not explain the stability and transport-wind separation.

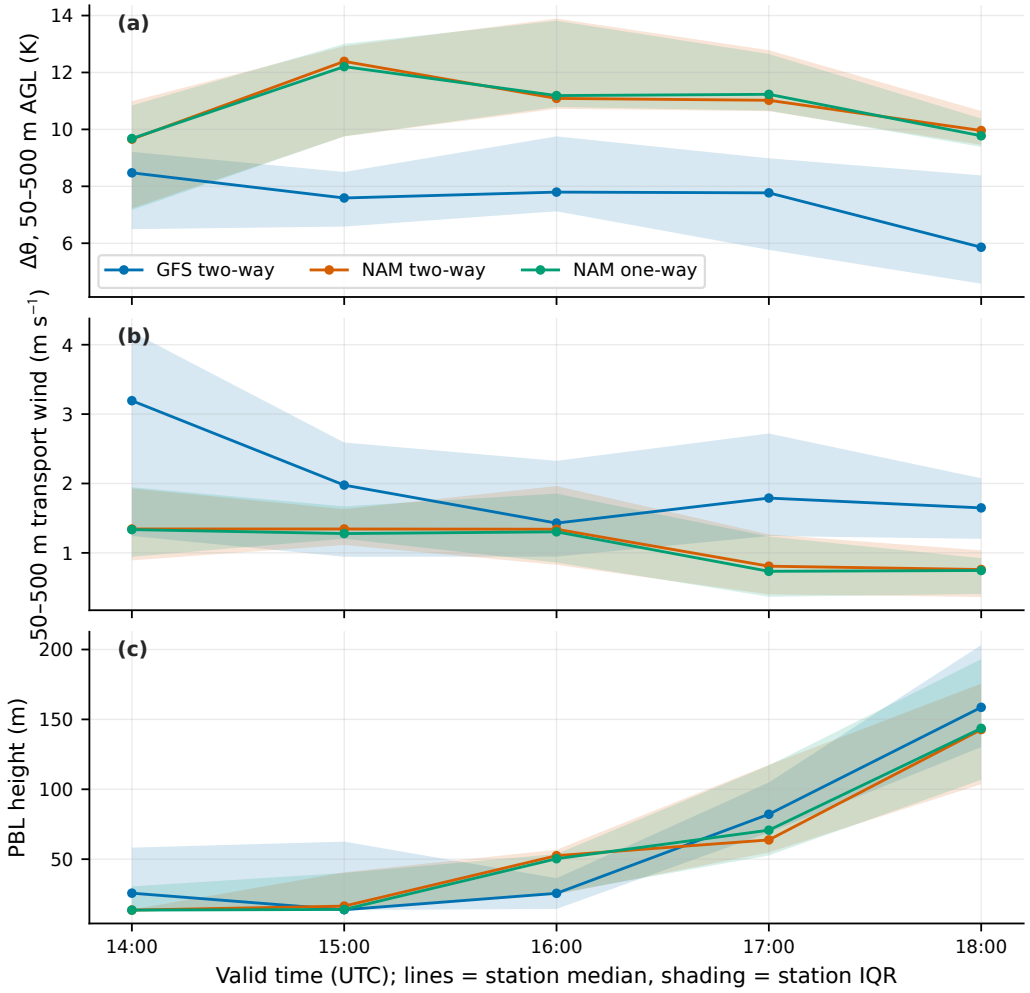


Figure 3. Time evolution of station-column medians and interquartile ranges for 50–500 m AGL potential-temperature stability, fixed-layer transport wind, and native WRF PBL height. GFS and NAM differ coherently in stability and transport wind, whereas the one-way and two-way NAM solutions nearly overlap.

3.3. Response to Nest Feedback

The controlled NAM one-way-minus-two-way response was small relative to the driving-analysis response. Its mean hourly station-median difference was +0.002 K for 50–500 m stability and +0.006 m s^{-1} for transport wind. The respective block sensitivity intervals were -0.039 to +0.053 K and -0.017 to +0.026 m s^{-1}. The surface-temperature-bias difference was -0.012 °C, with an interval spanning zero, while the 10 m-wind-bias difference was -0.038 m s^{-1}. These results bound the feedback response only for the present five-hour d02 sample and do not establish general equivalence between one-way and two-way nesting.

4. Discussion

The driving analysis was the dominant tested control on the short-window cold-pool structure. The GFS-driven solution combined a larger warm surface error, weaker 50–500 m stability, and stronger fixed-layer transport wind than the NAM-driven solution. Those differences describe the meteorological state supplied to a downstream chemistry model, but the present experiment cannot partition the response between the 12:00 UTC initial state and the subsequent lateral boundary conditions.

The near-null nest-feedback result narrows the interpretation of the primary comparison because the large GFS–NAM separation is not reproduced by changing feedback alone under NAM forcing. It does not show that feedback is unimportant for longer forecasts, the 333 m nest interior, different domain geometry, or other synoptic conditions. The current resolution ladder, nudging lineage, terrain experiments, and PBL treatments were therefore excluded from the first-draft Results rather than used as additional nominal sensitivities.

For air-chemistry applications, the relevant implication is that the GFS-driven run supplied a less stable and more strongly ventilated lower layer during the evaluated window. Such a state increases the atmospheric volume available for dilution and can hinder modeled near-surface accumulation, but it does not establish that meteorology is the only or dominant source of winter-ozone error. Emissions, deposition, photolysis, snow representation, and chemical mechanisms remain material uncertainties.

The principal limitation is temporal scope. Five post-spin-up hours can demonstrate a coherent structural difference but cannot establish cold-pool persistence across multiple diurnal cycles or the full ozone-accumulation period. The station columns also sample fixed locations rather than basin volume, and the observed bench-minus-floor contrast is horizontal rather than a colocated sounding.

5. Conclusions

For the matched 2 February 2013 simulations, the driving analysis materially affected the short-window cold-pool state, whereas changing NAM nest feedback produced only a very small response on the common d02 station sample. The GFS-driven simulation was warmer at the surface, less stable from 50 to 500 m AGL, and more strongly ventilated than the NAM-driven simulation. These findings provide a meteorological basis for later coupled modeling while remaining limited to a five-hour free-forecast comparison and leaving chemical and inventory uncertainties unresolved.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

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